LECTURE 04: NEURAL NETWORKS & GRADIENT DESCENT OPTIMIZATION
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Course: CS 482 - Deep Learning & Statistical Learning Theory

1. Introduction to Artificial Neural Networks (ANNs)
Artificial Neural Networks are computational models inspired by biological neural networks in the human brain. An ANN i

The mathematical formulation for a single neuron is given by:
z = (w_1 * x_1) + (w_2 * x_2) + ... + (w_n * x_n) + b = (W^T * X) + b
a = sigma(z)

Where:
- W represents the weight vector representing synaptic connection strengths.
- X represents the input feature vector.
- b represents the scalar bias term shifting the activation threshold.
- sigma represents the non-linear activation function (such as ReLU, Sigmoid, or Tanh).

2. The Role of Non-Linear Activation Functions
Without non-linear activation functions, a multi-layer neural network collapses into a single linear transformation, regardl
W_2 * (W_1 * X) = (W_2 * W_1) * X = W_combined * X

Common activation functions include:
- Rectified Linear Unit (ReLU): f(x) = max(0, x). ReLU mitigates the vanishing gradient problem in deep architectures, th
- Sigmoid Function: f(x) = 1 / (1 + exp(-x)). Maps inputs to the range (0, 1), often used in binary classification output laye
- Hyperbolic Tangent (Tanh): f(x) = (exp(x) - exp(-x)) / (exp(x) + exp(-x)). Zero-centered with output range (-1, 1).

CS 482: LECTURE 04 (CONTINUED) - LOSS FUNCTIONS & BACKPROPAGATION

3. Loss Functions and Empirical Risk Minimization
A loss function measures the discrepancy between the network's predicted output (y_hat) and the ground truth label (y).

- Mean Squared Error (MSE): Used predominantly for regression tasks.
  L_MSE = (1 / (2 * N)) * sum((y_i - y_hat_i)^2)

- Categorical Cross-Entropy Loss: Used for multi-class classification problems with Softmax output.
  L_CE = - sum(y_i * log(y_hat_i))

4. The Backpropagation Algorithm
Backpropagation is the foundational algorithm used to calculate the gradient of the loss function with respect to every in

The chain rule dictates that the gradient of the loss L with respect to an internal weight w_ij is:
dL / dw_ij = (dL / da_j) * (da_j / dz_j) * (dz_j / dw_ij)

By caching intermediate activation values during the forward pass, backpropagation propagates error derivatives backw

5. Gradient Descent Optimizers
Once gradients are computed, an optimization algorithm updates the parameters in the direction of steepest descent:
W_new = W_old - (eta * dL / dW)

Where eta represents the learning rate (step size):
- Stochastic Gradient Descent (SGD): Updates weights using a single training sample at a time, introducing high varianc
- Mini-Batch Gradient Descent: Evaluates gradients across a mini-batch of k samples (e.g., 32, 64, 128), balancing com
- Adam Optimizer (Adaptive Moment Estimation): Combines the benefits of momentum (exponential moving average of

CS 482: LECTURE 04 (CONTINUED) - REGULARIZATION & SUMMARY

6. Overfitting and Regularization Techniques
Overfitting occurs when a neural network memorizes noise in the training set rather than learning generalizable underlyi

Standard regularization strategies include:
- L2 Regularization (Weight Decay): Adds a penalty proportional to the sum of squared weights to the loss function:
  L_total = L_data + (lambda / 2) * sum(w_i^2)
  This discourages excessively large weights and encourages smoother decision boundaries.
- Dropout: During each training iteration, randomly deactivates a fraction p (e.g., p = 0.5) of hidden neurons. This prever
- Early Stopping: Halts training when validation loss stops decreasing and begins to increase, preventing over-training.

7. Key Exam Takeaways
- Understand why ReLU replaced Sigmoid in hidden layers (addressing vanishing gradients).
- Know the exact mathematical steps of the forward pass and backward chain rule.
- Differentiate between Batch Gradient Descent, Mini-batch SGD, and Adam optimization.
- Be able to derive the gradient of Mean Squared Error with respect to linear weights.
- Always check that your learning rate is neither too large (divergence) nor too small (premature stagnation).